Color Filters Basics

https://www.stagelightingstore.com

Color filters are tools or devices that selectively allow certain colors of light to pass through while blocking others. They are used in photography, filmmaking, lighting, and visual arts to alter the appearance of images or scenes.

Here's a simple breakdown:

- Purpose: To change or enhance the colors in an image or scene.

- How they work: They absorb some wavelengths of light and let others pass through.

- Examples: A red filter lets red light pass but blocks green and blue light, making colors appear
  more vivid or altering the mood of a photo.

- Uses: Creating special effects, correcting color balance, or emphasizing specific colors.

Since white light consists of red, blue, and green light, the following set of arrows will be used to represent white light.

White Light, Made from Red, Blue, and Green

In the following diagram we see that the cyan filter takes out the red and lets the blue and green pass through. Then the blue and green combine to form cyan. Note that in the simulation, only the green arrow and blue arrow will show. Leaving it up to you to know that together they form cyan.)

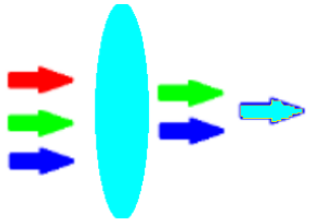


figure 1

In the following diagram we see that the magenta filter takes out the green and lets the blue and red pass through. Then the blue and red combine to form cyan.

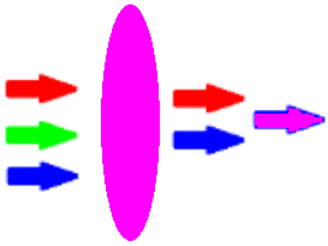


figure 2

In the following diagram we see that the yellow filter takes out the blue and lets the green and red pass through. Then the green and red combine to form cyan.

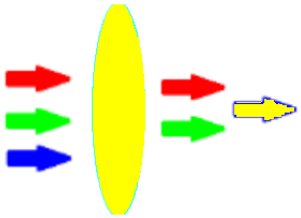
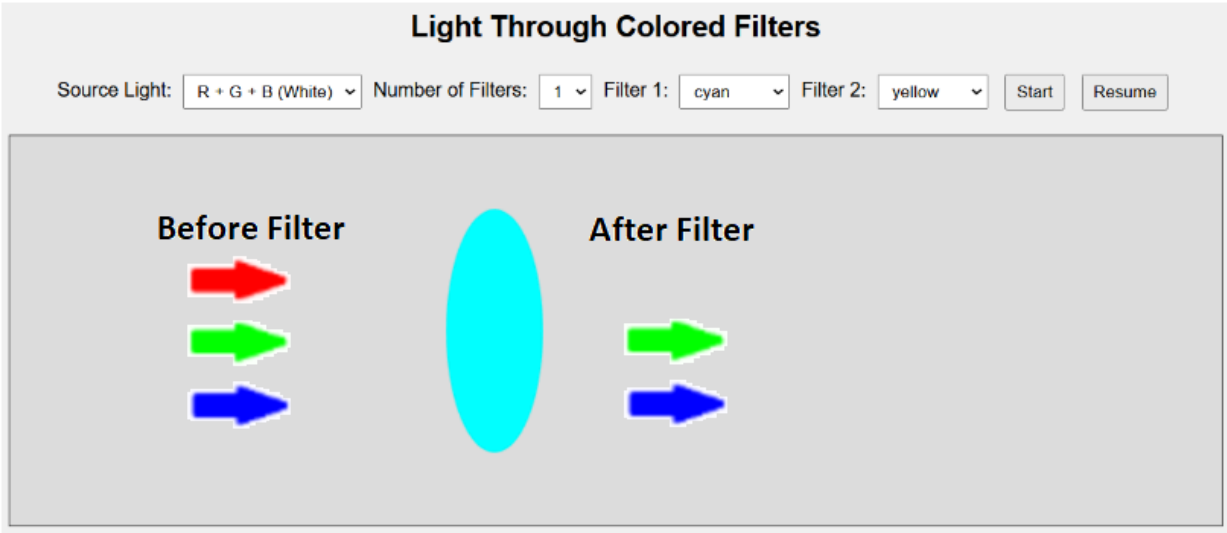


figure 3

As can be seen in the last three examples, the name of a filter is the same as the name of the light that passes through it: a cyan filter transmits cyan light, a magenta filter transmits magenta light, and a yellow filter transmits yellow light.

Below is a partial screenshot of the simulation. Use it to verify the results in *figures 1 - 3.*



Cyan Filter Takes Out Red Light

How to Use This Simulation

This simulation shows how red, green, and blue light pass through colored filters. Each arrow represents one component of light leaving the source.

When light reaches a filter, the filter only allows certain colors to pass through. Any colors that are not transmitted by the filter are absorbed and disappear.

Suggested Procedure

1. Select the source light (white, yellow, magenta, etc.).
2. Select one or two filters.
   (If you choose 1 filter, you can ignore the color of the filter showing in the dropdown for
   filter 2. The app ignores it.)
3. Press **Start** . (Your choices won't display until you click Start..
4. Watch the arrows move toward the filters.

Using the Pause Button

A useful strategy is to press **Pause** just before the light reaches a filter. This gives you time to predict what will happen next.

Ask yourself:

- Which colors does this filter allow to pass?
- Which colors will be absorbed?
- What color light should remain after the filter?

Then press **Resume** to see if your prediction is correct.

Try experimenting with different combinations of light sources and filters to see how the colors change.

More Examples (Be sure to verify each result using the app.)

Example 1: Red filters let red light pass through, therefore they absorb green and blue:

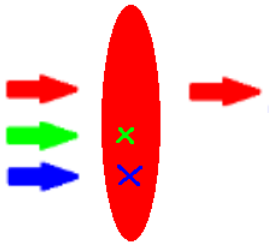


figure 4

Example 2: Green filters let green light pass through, they must absorb red and blue.

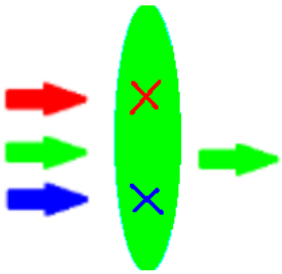


figure 5

Example 3: Blue filters let blue light pass through, so they must absorb red and green.

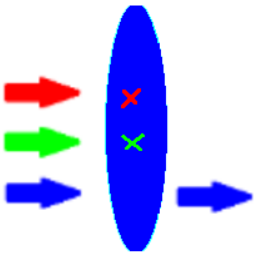


figure 6

Example 4:

Green, blue, and red light impinge on a magenta filter. Which of these primary colors are transmitted through the filter, and how does the human brain interpret these results?

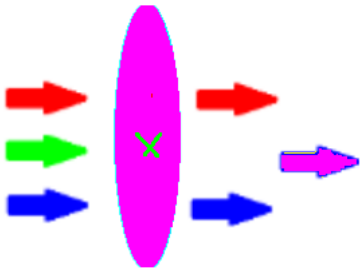


figure 7

The magenta filter absorbs the green light and lets the red and blue pass through, which the brain interprets as magenta.

Example 5:

White light impinges on a yellow filter. Then the light transmitted through the yellow filter impinges on a magenta filter. Then the remaining transmitted light illuminates a screen. What is the color of the screen?

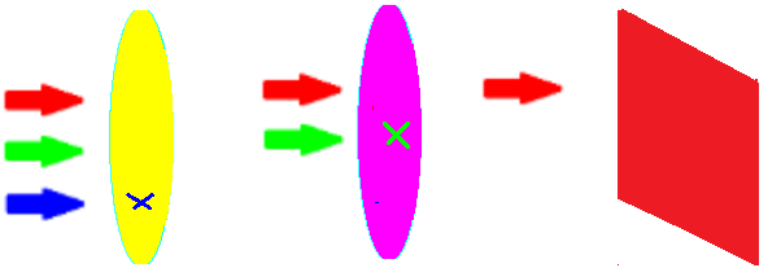


figure 8

From left to right, the diagram shows white rgb light impinging on a yellow filter which takes out the blue from the rgb, and allows the red and green to transmit through it. Then the red and green impinge on the magenta filter, which takes out the green light and allows the red component to pass through where it then illuminates the screen.

Example 6:

Cyan light impinges on a cyan filter. Then the light transmitted through the cyan filter impinges on a yellow filter. Then the remaining transmitted light illuminates a screen. What is the color of the screen?

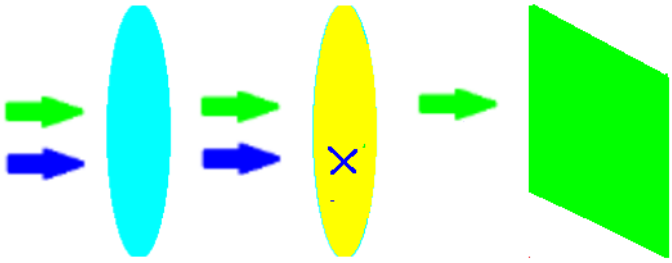


figure 9

From left to right, the diagram shows cyan (gb) light impinging on a cyan filter, which doesn't take out either the green or the blue, because cyan filters only absorb red light. So the cyan (gb) light passes through the cyan filter and impinges on the yellow filter. The yellow filter absorbs the blue and allows the green light to transmit through it, after which it impinges upon and illuminates the screen.

This table tells students **which colors pass through each filter** and **which are absorbed**.

Filter Color	**Passes (Transmits)**	**Absorbs (Arrows)**
Red	Red	Green, Blue
Green	Green	Red, Blue
Blue	Blue	Red, Green
Cyan	Green, Blue	Red
Magenta	Red, Blue	Green
Yellow	Red, Green	Blue